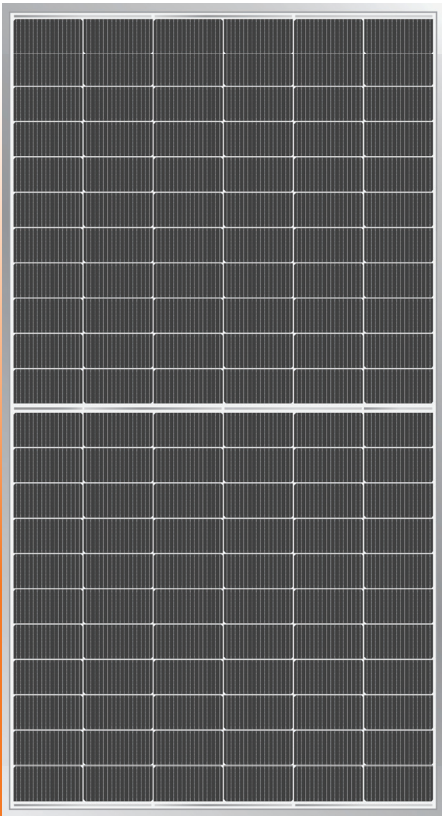




G12R TOPCon Module Datasheet

ISL132NG12R

600-625W
132 Half-Cut Cells



Certifications



12-Year Material
Warranty



30-Year Linear
Power Warranty

625W
MAXIMUM
POWER OUTPUT

23.1%
MAXIMUM
EFFICIENCY

0~+5W
POSITIVE POWER
TOLERANCE

UPTO
80%
BIFACIALITY
FACTOR

HIGHLIGHTS



Up to 30% More Power Output
Delivers significantly higher energy
yield compared to conventional
P-type modules.



Superior Low-Light Performance
Generates more power even in low-
irradiance conditions such as cloudy
or foggy weather.

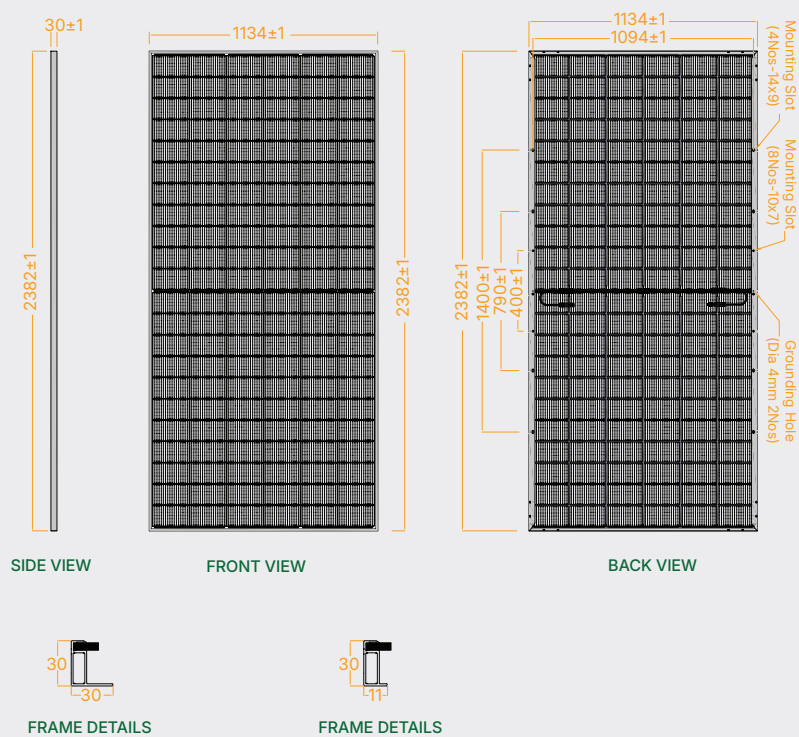


Enhanced Temperature Tolerance
Maintains high power output even in
high ambient temperatures, thanks to
a better temperature coefficient.

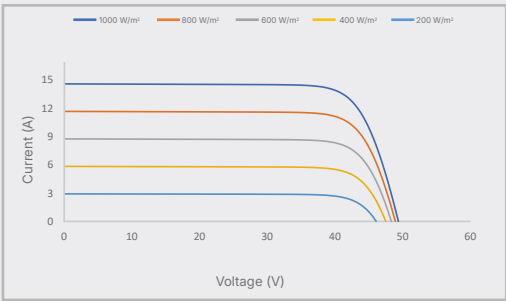


**Zero LID (Light-Induced
Degradation)**
Ensures consistently high energy
generation without initial power loss.

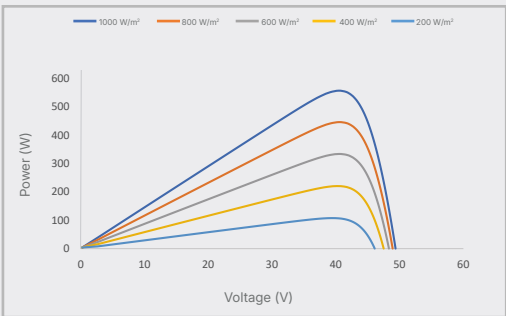
DIMENSIONS OF PV MODULE(MM)



IV CURVE



PV CURVE

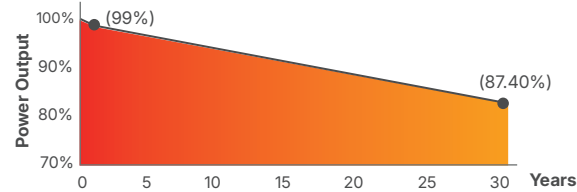


ELECTRICAL PARAMETERS

Module Type	ISL NG12R 600		ISL NG12R 605		ISL NG12R 610		ISL NG12R 615		ISL NG12R 620		ISL NG12R 625	
Power Tolerance - P _{MAX} (W)	0 - +5w											
Parameters	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT	STC	NOCT
Maximum Power (Pmax)	600	458	605	461	610	465	615	469	620	473	625	477
Open Circuit Voltage (Voc)	48.75	46.53	48.85	46.62	48.94	46.71	49.05	46.82	49.15	46.91	49.25	47.01
Short Circuit Current (Isc)	15.56	12.44	15.66	12.52	15.76	12.6	15.86	12.68	15.96	12.76	16.06	12.84
Maximum Power Voltage (Vmp)	40.69	38.84	40.78	38.92	40.85	38.98	40.96	39.09	41.05	39.18	41.14	39.26
Maximum Power Current (Imp)	14.76	11.8	14.85	11.87	14.95	11.95	15.03	12.01	15.12	12.08	15.21	12.16
Module Efficiency -ηm(%)	22.2		22.4		22.6		22.8		23		23.1	

STC: Irradiance: 1000w/m², Cell Temperature : 25°C, Atmospheric=1.5 | NOCT: Irradiance: 800w/m², Cell Temperature: 45°±2C, Atmospheric = 1.5, Wind speed :1m/s

30-YEAR LINEAR POWER WARRANTY



OPERATION PARAMETERS

Operational Temperature	-40°C~+85°C
Maximum System Voltage	1500VDC
Maximum Series Fuse Rating	30A
Power Tolerance	0~+5W
Nominal Operating Cell Temperature(NOCT)	42°C ± 2°C
Safety Class	Class II
Refer. Bifacial Factor	80+ -10%

SPECIFICATIONS

Cells Type	TOPCon, 16BB
No. of cells & Matrix	132 [(11×6)x2]
Module Dimensions [mm]	2382(L) × 1134(W) × 30(T)
Weight	33.0 Kg
Front Glass Thickness	2 mm
Rear Glass Thickness	2 mm
Junction Box & Cable Length	IP68, 300/400mm

LOAD CAPACITY

Maximum Load Front (pa)	5400
Maximum Load Rear (pa)	2400
Through Hail Testing	25mm diameter, Impact Speed 23m/s

TEMPERATURE COEFFICIENT (STC TEST)

Temperature Coefficient of P _{MAX}	-0.28%/°C
Temperature Coefficient of V _{oc}	- 0.24%/°C
Temperature Coefficient of I _{sc}	+0.04%/°C

WARRANTY

12-Year Material Warranty
30-Year Linear Power Warranty
1% First Year Degradation
0.40% Annual Power Attenuation

PACKING CONFIGURATION

Container Capacity 40' HC
Modules Per Pallet: 36 Pieces
Pallets Per Container: 20 Pallets
Total Modules Per Container: 720 Pieces

CAUTION:
READ SAFETY AND INSTALLATION INSTRUCTIONS BEFORE USING THE PRODUCT.

Document Number: DS/ISL132NG12R/02